Physics 468/668: Statistical Physics **Spring 2019**

Lecture 10: Harmonic Oscillator

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10.1 Harmonic oscillators

A system of N independent harmonic oscillators is representing two important problems,

1. the theory of black-body radiation
2. the theory of lattice vibrations

We start with the specialized situation when the oscillators can be treated classically. The Hamiltonian of any one of them is then given by

$$H(q_i, p_i) = \frac{1}{2}m\omega^2 q_i^2 + \frac{1}{2m}p_i^2 \tag{10.1}$$

For the single-oscillator partition function, we readily obtain

$$\begin{aligned} Q_1(\beta) &= \int\int \exp[-\beta(\frac{1}{2}m\omega^2 q^2 + \frac{1}{2m}p^2)]\frac{dqdp}{h} \\ &= \frac{1}{h}\left(\frac{2\pi}{\beta m\omega^2}\right)^{1/2}\left(\frac{2\pi m}{\beta}\right)^{1/2} \\ &= \frac{1}{\beta\hbar\omega} = \frac{kT}{\hbar\omega} \end{aligned} \tag{10.2}$$

This representing a classical counting of the average number of accessible micro states - that is, kT divided by the quantum harmonic oscillator energy spacing. The partition function of the $N-$oscillator system would be

$$Q_N(\beta) = [Q_1(\beta)]^N = (\beta\hbar\omega)^{-N} = \left(\frac{kT}{\hbar\omega}\right)^N \tag{10.3}$$

here we assume that the oscillators are distinguishable. It is actually photons in one case and phonons in the other, which distribute themselves over the various oscillator levels, that are indistinguishable.

A is now given by

$$A = -kT\ln Q_N = NkT\ln\frac{\hbar\omega}{kT} \tag{10.4}$$

And

$$\mu = kT\ln\left(\frac{\hbar\omega}{kT}\right) \tag{10.5}$$

$$P = 0 \tag{10.6}$$

$$S = Nk\left[\ln(\frac{\hbar\omega}{kT}) + 1\right] \tag{10.7}$$

$$U = NkT \tag{10.8}$$

$$C_p = C_V = Nk \tag{10.9}$$

We can also determine the density of states $g(E)$

$$g(E) = \frac{1}{(\hbar\omega)^N}\frac{1}{2\pi i}\int_{\beta'+i\infty}^{\beta'-i\infty}\frac{e^{(\beta E)}}{\beta^N}d\beta \tag{10.10}$$

$$g(E) = \begin{cases} \frac{1}{(\hbar\omega)^N}\frac{E^N-1}{(N-1)!} & \text{if } E \geq 0 \\ 0 & \text{if } E < 0 \end{cases} \tag{10.11}$$

Taking $N \ll 1$ and using the Stirling approximation, we get

$$S(N,E) = k\ln g(E) \approx NK\left[\ln\left(\frac{E}{N\hbar\omega}\right) + 1\right] \tag{10.12}$$

which gives for the temperature of the system

$$T = \left(\frac{\partial S}{\partial E}\right)_N^{-1} = \frac{E}{Nk} \tag{10.13}$$

10.2 Quantum situation

When it comes to the quantum situation,

$$\epsilon_n = (n + 1/2)\hbar\omega \tag{10.14}$$

We have for the single-oscillator partition function

$$Q_1(\beta) = \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega} = \frac{\exp(-1/2\beta\hbar\omega)}{1-\exp(-\beta\hbar\omega)} \tag{10.15}$$

The N-oscillator partition function is then given by

$$Q_N(\beta) = [Q_1(\beta)]^N = \frac{\exp(-N/2\beta\hbar\omega)}{(1-\exp(-\beta\hbar\omega))^N} \tag{10.16}$$

Therefore, we get

$$A = NkT\ln Q_1(\beta) = NkT\ln\left[\frac{\exp(-1/2\beta\hbar\omega)}{(1-\exp(\beta\hbar\omega))}\right] = N\left[\frac{1}{2}\hbar\omega + kT\ln(1-e^{-\beta\hbar\omega})\right] \tag{10.17}$$

$$\mu = A/N = \frac{1}{2}\hbar\omega + kT\ln(1 - e^{-\beta\hbar\omega}) \tag{10.18}$$

$$P = 0 \tag{10.19}$$

$$S = Nk\left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega} - 1} - \ln(1 - e^{-\beta\hbar\omega})\right] \tag{10.20}$$

$$U = N\left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega - 1}}\right] \tag{10.21}$$

$$C_P = C_V = Nk(\beta\hbar\omega)^2\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2} \tag{10.22}$$

Clearly, the quantum mechanical oscillators do not obey the equipartition theorem. The mean energy per oscillator is different from kT. Only in the high T limit, the mean energy tend to be equal to kT.

For $kT \gg \hbar\omega$, the formula goes to classical picture.

We shall now determine the density of states $g(E)$ of the N-oscillator system from its partition function. Carrying out the binomial expansion of the expression, we have

$$Q_N(\beta) = \sum_{R=0}^{\infty}\binom{N+R-1}{R}e^{-\beta(\frac{1}{2}N\hbar\omega + R\hbar\omega)} \tag{10.23}$$

Comparing this with the formula

$$Q_N(\beta) = \int_0^{\infty} g(E)e^{-\beta E}dE \tag{10.24}$$

We conclude that

$$g(E) = \sum_{R=0}^{\infty}\binom{N+R-1}{R}\delta\left(E - [R + \frac{1}{2}N]\hbar\omega\right) \tag{10.25}$$

where $\delta(x)$ denotes the Dirac delta function. It implies that there are $(N+R-1)!/R!(N-1)!$ microstates available to the system when its energy E has the discrete value $(R + \frac{1}{2}N)\hbar\omega$, where R= 0,1,2,..., and that no micro-state is available for other values of E. This is hardly surprising, but it is instructive to look at this result from a different point of view.

We consider the problem that arises naturally in the micro-canonical ensemble theory. Given an energy E for distribution among a set of N harmonic oscillators, each of which can be in any one of the eigenstates, what is the total number of distinct ways in which the process of distribution can be carried out? In view of the form of the eigenvalues ϵ_n, it makes sense to give away, right in the beginning, the zero-point energy $\frac{1}{2}\hbar\omega$ to each of the N oscillators and convert the rest of it into quanta. Let R be the number of these quanta, then

$$R = \left(E - \frac{1}{2}N\hbar\omega\right)/\hbar\omega \tag{10.26}$$

Clearly, R must be an integer, by implication, E must be of the form $(R + \frac{1}{2}N)\hbar\omega$. The problem then reduces to determining the number of distinct ways of arranging R to N oscillators. The answer is clearly

$$\frac{(R+N-1)!}{R!(N-1)!} \tag{10.27}$$



Figure 10.1: Distribution of energies.

We can now determine the entropy of the system from the number, since $N \gg 1$,we have,

$$\begin{aligned} S &\approx k[\ln(R+N)! - \ln R! - \ln N!] \\ &\approx k[(R+N)\ln(R+N) - R\ln R - N\ln N] \end{aligned} \tag{10.28}$$

The number R is a measure of the energy E fo the system, we can calculate,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N = \left(\frac{\partial S}{\partial R}\right)_N \frac{1}{\hbar\omega} = \frac{k}{\hbar\omega}\ln\left(\frac{R+N}{R}\right) = \frac{k}{\hbar\omega}\ln\left(\frac{E+\frac{1}{2}N\hbar\omega}{E-\frac{1}{2}N\hbar\omega}\right) \tag{10.29}$$

so that,

$$\frac{E}{N} == \frac{1}{2}\hbar\omega\frac{\exp(\hbar\omega/kT)+1}{\exp(\hbar\omega/kT)-1} \tag{10.30}$$

This is consistent with our thermodynamic limit.

Finally, we may consider

$$\frac{(R+N-1)(R+N-2)...(R+1)}{(N-1)!} \approx \frac{R^N-1}{(N-1)!} \tag{10.31}$$

with

$$R \approx E/\hbar\omega \tag{10.32}$$

The corresponding expression for the entropy turns out to be

$$S \approx k[N\ln(R/N)+N] \approx Nk[\ln(\frac{E}{N\hbar\omega})+1] \tag{10.33}$$

which gives

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N \approx \frac{Nk}{E} \tag{10.34}$$

so

$$\frac{E}{N} \approx kT \tag{10.35}$$